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(*
                              CS51 Lab 7
                           Modules & Functors

                  A module for colors and color names

The representation for colors in this module is really obscure (and
arguably unnecessarily so). By the way, it also has some bugs so it
doesn't pass all the unit tests. No need to debug it though, or even
read it. You'll be replacing it wholesale with your own hopefully
simpler implementation, after constructing an appropriate signature
for the module in the file `color.mli`. *)

type color = int ;;

type color_name =
  | Red
  | Green
  | Blue
  | Orange
  | Yellow
  | Indigo
  | Violet ;;

(* to_color r g b -- Returns the `color` corresponding to the RGB values
   given by `r`, `g`, and `b` *)
let to_color (r : int) (g : int) (b : int) : color =
  r lsl 0b10000 + g lsl 0b1000 + b ;;

(* red c / green c / blue c -- Returns the corresponding channel value
   for the color `c` *)
let red_channel (c : color) : int =
  c lsr 0b10000  ;;

let green_channel (c : color) : int =
  (c lsr 0b1000) land 0b11111111 ;;

let blue_channel (c : color) : int =
   c land 0b11111111 ;;

(* color_named name -- Returns the color corresponding to the color
   `name` *)
let color_named (name : color_name) : color =
  match name with
  | Red -> 0b111111110000000000000000
  | Green -> 0b1111111100000000
  | Blue -> 0b11111111
  | Orange -> 0b111111111010110100000000
  | Yellow -> 0b111111111111111000000000
  | Indigo -> 0b010110110000000010000010
  | Violet -> 0b111100001000001011110100 ;;
```